Math 54 Discussion Worksheet

Name: ____

Problem 1

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$$

be a 3×3 matrix.

(a) Compute $\det(A)$.

(b) Without recomputing from scratch, compute $\det(B)$ where B is obtained from A by replacing the third row with $(\text{row}_3) - 2(\text{row}_1)$.

(c) Use your answers to decide whether A is invertible.

Problem 2

a) Suppose that A is a square matrix such that $\det(A^3) = 0$. Explain why A cannot be invertible.

b) Let A and B be square matrices. Show that even though AB and BA may not be equal, it is always true that $\det(AB) = \det(BA)$.

c) Let A and P be square matrices, with P invertible. Show that $\det(PAP^{-1}) = \det(A)$.

d) Let U be a square matrix such that $U^T U = I$. Show that $\det(U) = \pm 1$.

e) Find a formula for $\det(rA)$ when A is an $n \times n$ matrix.

Problem 3

Suppose T and U are linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^n$ such that $T(U(x)) = x$ for all $x \in \mathbb{R}^n$. Is it true that $U(T(x)) = x$ for all $x \in \mathbb{R}^n$? Why or why not?